SMITH ENGINEERING

Queen's University

Department of Electrical and Computer Engineering



ELEC 490/498 Project Blueprint

Autonomous Smart Caddy Cart (ASCC)

Fall 2026 / Winter 2027

Submitted by Group 07

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Submission date: November 4th, 2026

Version: 01

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Acronyms

AoA angle of arrival
ASCC Autonomous Smart Caddy Cart
BMS battery management system
DC-DC direct current to direct current
FOC field-oriented control
GPIO general-purpose input/output
GPS Global Positioning System
I2C Inter-Integrated Circuit
IP ingress protection
LiDAR light detection and ranging
MVP minimum viable product
NFC near-field communication
PoC proof of concept
RFID radio-frequency identification
RTK real-time kinematic
SPI Serial Peripheral Interface
UART universal asynchronous receiver-transmitter
UWB ultra-wideband

1. Introduction

1.1 Motivation and Problem

Golfers who walk the course must transport their equipment by carrying a bag or operating a push cart. Although electric and autonomous carts reduce this effort, they primarily focus on transportation rather than equipment management. Golfers must also keep track of clubs removed during play, which can be misplaced or left behind.

The Autonomous Smart Caddy Cart (ASCC) aims to address both challenges by combining autonomous golfer following with club identification and tracking. As golfers ourselves, the team understands the need for a solution that addresses both transportation and equipment management. The goal is to reduce manual cart operation while helping golfers manage their equipment throughout a round.

1.2 Project Objectives

The ASCC will be developed as a motorized golf push cart capable of following a golfer using a controller-based tracking system rather than computer vision. Distance sensors will support obstacle detection and avoidance, while an emergency-stop mechanism will allow the cart to be stopped when necessary.

An RFID-based subsystem will identify tagged golf clubs and monitor their presence in the bag, notifying the golfer when a club is missing. The objective is to integrate these functions into a reliable and safe prototype.

The ASCC will also aim to incorporate golf course data, including hole layouts, distances, and course information, to assist golfers with shot planning. This information helps golfers estimate distances to targets and hazards, supporting more informed club selection and on-course decisions.

1.3 Project Scope

The minimum viable prototype will focus on motor control, golfer following, obstacle detection, emergency stopping, RFID-based club tracking, and basic golf course data integration. Additional features, including shot data collection, club recommendations, and speech functionalities, may be implemented if time and resources permit.

Testing will begin indoors before progressing to controlled outdoor environments. The prototype is not intended for commercial deployment or operation on public roads, crowded areas, or extreme terrain.

1.4 System Overview and Operating Scenario

Figure 1 illustrates the ASCC's main subsystems and their interactions with the golfer and surrounding environment. The Raspberry Pi coordinates sensor inputs, motor control, and club tracking.

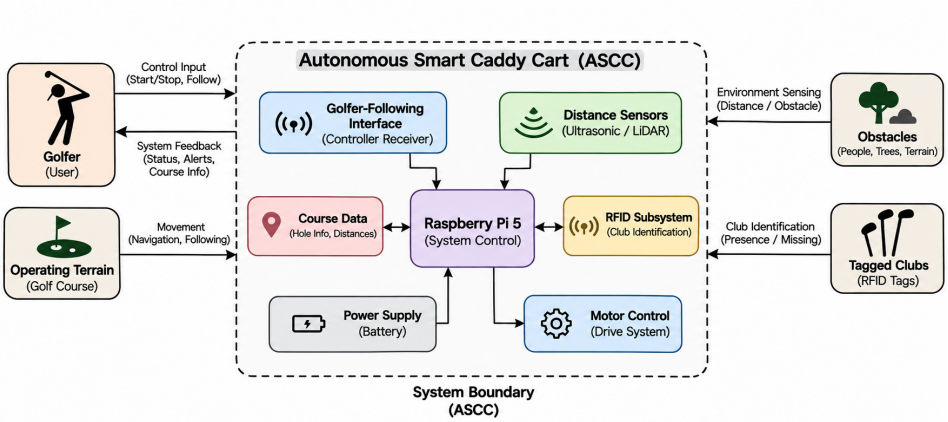


Figure 1: ASCC system overview and external interactions.

1.5 Design Constraints

The ASCC must be completed within the CAD $600 budget and capstone development schedule. Existing components, including the Raspberry Pi, ultrasonic sensors, and golf push cart, will be reused to reduce costs.

Key constraints include hardware compatibility, sensor reliability, battery capacity, and safe motor operation. These considerations will guide the system requirements, development approach, and verification procedures.

2. Functional Description

2.1 System Overview and Design Alternatives

2.2 Functional Decomposition

2.3 Logical Decomposition and Interfaces

2.4 Physical Architecture

2.5 Prototype Definitions

3. Requirements Specifications

3.1 Stakeholder Needs and Requirements Hierarchy

The ASCC requirements were developed around five stakeholder needs: equipment transportation, safety, club management, course information, and reliable operation. These address limitations of conventional motorized golf carts, which still require user control and do not assist with tracking equipment. The needs were decomposed into system-level and subsystem requirements to ensure each technical function contributes to the overall project objectives. Safety and essential movement functions were prioritized due to the risks associated with autonomous operation and the project's limited development resources. The requirements hierarchy is provided in Tables 7 and 8.

3.2 Functional Requirements

The primary function is autonomous golfer following, eliminating the need for continuous manual control. UWB tracking was selected to estimate the golfer's relative position without the computational demands and lighting dependencies of computer vision. Obstacle detection and emergency stopping provide safeguards when following commands could result in unsafe movement. RFID club tracking addresses the risk of forgotten equipment, while GPS-based course information improves convenience by displaying relevant distances and hole information. These functions were separated into independently testable requirements to support incremental development and integration, as detailed in Table 9.

3.3 Non-Functional and Interface Requirements

Performance targets were selected to balance usability, safety, and prototype feasibility. The 1–3 m following distance keeps the cart accessible while providing separation from the golfer. A target speed of 2 m/s allows the cart to keep pace with brisk walking, although

normal operating speeds may be lower to maintain stability and reduce stopping distance. The initial 1 m obstacle detection range provides a measurable sensor benchmark, but the final stopping zone must account for braking distance, control latency, and sensor uncertainty.

The 90% RFID accuracy target establishes an initial reliability benchmark while accounting for tag orientation and reader placement. A 1-hour battery runtime supports extended testing without frequent recharging. These preliminary targets will be validated experimentally, as outlined in Table 10. Interface requirements define communication and power exchange between subsystems, enabling independent development and reliable integration (Table 11).

3.4 Safety and Fault Requirements

Safety requirements prioritize preventing uncontrolled movement during emergency stops, tracking loss, and sensor faults. Emergency stopping must override autonomous commands and require manual reset to prevent unexpected restarting. A preliminary 1 s tracking timeout limits movement based on outdated position information, while invalid obstacle measurements must result in a safe stopped state rather than continued operation. These thresholds will be evaluated through braking tests and fault injection before outdoor testing. Tables 12 and 15 provide the detailed requirements and verification procedures.

3.5 Critical Requirements and Traceability

Requirements were classified as critical, major, or minor based on their importance to safety and prototype functionality. Critical requirements, particularly motor inhibition and obstacle response, must be verified before integrated autonomous testing. Traceability links each requirement to its stakeholder need, subsystem, prototype, and verification test, ensuring design decisions can be evaluated and changes tracked. For example, MC-04 requires motor actuation to be inhibited during an emergency stop and is verified through VT-07. The complete relationships are documented in Tables 13, 14, and 16.

3.6 Stretch Goals

Stretch goals include shot tracking, club recommendations, and voice-based interaction. These features could improve convenience and provide additional golfing insights but depend on reliable core subsystems. Voice recognition also introduces challenges from

outdoor noise and cannot replace the dedicated emergency-stop mechanism. These features are therefore excluded from mandatory prototype acceptance criteria and will only be pursued if time and resources permit (Table 17).

4. Risks and Safety Hazards

4.1 Hazard Identification and Risk Assessment

Hazards are assessed with the ISO 12100 process [1]: identification, risk estimation and evaluation, then risk reduction by inherently safe design (D), safeguarding and complementary protective measures (S), and information for use (I). Risk is scored with the MIL-STD-882E matrix [2], with severity from 1 (catastrophic) to 4 (negligible) and probability from A (frequent) to E (improbable), giving a level of High, Serious, Medium, or Low. The scope is the cart during testing and the demonstration. Technical and schedule risks are handled as contingencies in Table 5. Table 1 rates each hazard before mitigation, and Figure 2 in Appendix B shows the causes and controls for loss of motion control.

H-09 needs some golf context. A golfer takes a club out of the bag only to hit a shot, then stands beside the ball and swings the club in an arc around their body. The arc's radius is about arm length plus club length, and clubs other than putters may be up to 48 in (1.22 m) long [12]. With an arm length of about 0.7 m, the reach is about 1.9 m, so the proposal's 1.5 m following distance put the cart inside the arc.

Speed sets the size of H-01 and H-03. For the cart to stop within a gap $d_{\min}$ after a reaction time t_r, it must decelerate at

$$a_{\text{req}} = \frac{v^2}{2\,(d_{\min} - v\,t_r)}. \tag{1}$$

The proposal's worst case ($v = 5.5$ km/h, $t_r = 300$ ms, $d_{\min} = 1.2$ m) needs $a_{\text{req}} \approx 1.6$ m/s^2, and an estimated 30 kg [MEAS: loaded mass] cart then carries $E = \tfrac{1}{2}mv^2 \approx 35$ J. Stopping distance and impact energy both grow with v^2, so the speed cap is the first mitigation. The new values ([REQ: max speed], [REQ: reaction time], [REQ: d_min]) set a braking-deceleration requirement verified at T-DRV-4 (contingency C-9).

4.2 Risk Mitigation and Residual Risk

Table 2 lists the mitigations for each hazard by ISO 12100 step, the requirement each one produces, and the test that verifies it. Each *shall not* statement addresses a specific fault.

Table 1: Safety hazards and initial risk (before mitigation), scored as severity/probability with MIL-STD-882E.

ID	Hazard and ASCC-specific cause	Standard	Magnitude	Initial
H-01	**Cart strikes the golfer.** Golfer stops suddenly, or walks back to the bag to pull a club and ends up inside the following distance	ISO 13482 [3]; ISO 3691-4 [4]	Stopping distance $vt_r + v^2/2a$ vs. gap; impact energy $\frac{1}{2}mv^2$ (worked below)	3C Medium
H-02	**Cart strikes an undetected person or object.** Ultrasonic beam angle and mounting height leave a blind zone for low objects (sprinkler heads, curbs, a crouching person); grass causes false returns	ISO 13482	Blind zone from beam angle and height [MEAS: sensor height]; detection range must exceed stopping distance plus margin	3C Medium
H-03	**Uncommanded or runaway motion.** UWB tag dropped or handed to someone else; corrupted UART command or Pi crash; firmware speed limit misconfigured on a board capable of well above walking pace	ISO 13482	Travel before stop $= v_{\max} t_{\text{timeout}}$ ([REQ: max speed], [REQ: timeout]); uncapped speed [MEAS: stock top speed]	2C Serious
H-04	**E-stop fails or is too slow.** Radio link drops; stop routed through a hung Pi; cart restarts on reset	ISO 13850 [5]; IEC 60204-1 [6]	Travel $= v \times$ latency ([REQ: e-stop latency], [REQ: link-loss timeout])	2D Medium
H-05	**Rollaway or tip-over.** Power cut on a slope lets the unpowered hub motors freewheel; turning on a side slope with a tall loaded bag	ISO 13482	Rollaway force $mg \sin\theta$; tip angle $\arctan(w/2h)$ for track [MEAS: w], centre of gravity [MEAS: h], vs. site slope [MEAS: slope]	3C Medium
H-06	**Pinch or entanglement.** Exposed hub wheels beside the frame and caster catch hands, grips, or bag straps	ISO 13857 [7]; ISO 13854 [8]	Measured gaps [MEAS: gaps] vs. minimum gaps for fingers and hands	3D Medium
H-07	**Battery fire, burn, or shock.** 36 V lithium-ion pack; new wiring to the DC-DC converter and relay; short circuit, overload, charging	UL 2272 [9]; IEC 62133-2 [10]	Stored energy ≈ 158 Wh for a 36 V, 4.4 Ah pack [MEAS: pack label]; fuse rating vs. wire ampacity	1E Medium
H-08	**Water reaches the electronics.** Morning dew on grass; light rain during an outdoor test	IEC 60529 [11]	Enclosure IP rating vs. required [REQ: IP rating]	3D Medium
H-09	**Golfer's swing and the cart.** A golfer with a club out is about to swing; a cart within reach is struck or moves into the golfer mid-swing	Equipment Rules [12]	Reach $\approx$ arm + club $\approx 0.7 + 1.22 \approx 1.9$ m	3C Medium

Table 2: Mitigations (D: inherently safe design; S: safeguarding; I: information for use), derived requirements, verification tests, and residual risk.

ID	Mitigations	Derived requirement	Test	Residual
H-01	D: firmware speed cap; following distance set from Eq. (1). S: no approach when range $<$ $d_{\min}$; firmware brake to standstill	Shall not command motion toward the golfer when range $<$ [REQ: d_min]; shall decelerate at $\geq$ [REQ: a_req]	T-DRV-4, T-TRK-3	3E Medium
H-02	D: LiDAR plus downward-angled ultrasonic sensors covering low objects. S: obstacle stop. I: soft obstacles only in testing	Shall stop within [REQ: reaction time] of an obstacle at $\geq$ [REQ: detection range]	T-SAF-1, T-SAF-2	3D Medium
H-03	D: speed and current caps on the mainboard, independent of the Pi. S: serial-timeout stop; tag-loss timeout; reject implausible range jumps; Pi watchdog	Shall not exceed [REQ: max speed] under any command; shall not move more than [REQ: timeout] after loss of the tag or UART link	T-DRV-2, T-TRK-3, T-TRK-4	3E Medium
H-04	S: Category 1 stop (below); link loss counts as a stop; reset does not restart motion [5]; manually actuated device held by the safety operator	Shall not resume motion automatically after reset; shall stop within [REQ: e-stop latency] at up to [REQ: e-stop range]	T-SAF-3	2E Medium
H-05	D: low bag cradle; widest track the chassis allows. S: firmware standstill hold while powered. I: slopes $\leq$ [REQ: max slope] (tilt sensing is STR-11)	Shall not be operated on slopes above [REQ: max slope]	T-DRV-4 (hold on slope); tip calculation	3D Medium
H-06	D: guards over the hub wheels. I: kill switch off before any adjustment	Guard gaps shall meet ISO 13857	T-INS-1	4D Low
H-07	D: stock pack and BMS unmodified; inline fuse sized to the wiring; strain relief. I: original charger, off the cart, supervised	Accessory circuit shall be fused at [REQ: fuse rating]	T-INS-2; current at INT-1	1E Medium
H-08	D: rated enclosure. I: dry weather only; stop testing if rain starts	Enclosure shall meet [REQ: IP rating]	T-INS-3	3E Medium
H-09	S: club-out interlock blocks motion toward the golfer until the club returns. I: park the cart outside swing reach	Shall not move toward the golfer while a club is out of the bag	T-SAF-4	3E Medium

The e-stop uses a Category 1 stop [6]: the e-stop signal commands the firmware brake and, after a short delay, a relay removes motor power independent of the Pi. Cutting power immediately (Category 0) would let the hub motors freewheel, and a firmware-only stop depends on the Pi and the UART link. With both, a hung Pi still results in power removal, and on flat ground the cart has braked before power is cut.

The remaining risk is accepted for a supervised prototype because every test in which the cart moves has a safety operator holding the e-stop, and operation is limited to dry

weather and flat ground. Four residual risks remain and are stated as limitations in Section 8. Unpowered hub motors freewheel, so the cart can roll on a slope after power is cut (H-05); shorting the motor phases through a relay would add braking if time allows. The sensors can still miss thin or very low objects (H-02). A battery fire stays catastrophic; only its probability is reduced (H-07), as the 2016 hoverboard recalls for fire hazard show [13]. The golfer can ignore the parking guidance and swing near a stopped cart (H-09).

5. Approach

5.1 Development Strategy and Division of Labour

The Autonomous Smart Caddy Cart (ASCC) is developed incrementally on a used hoverboard whose mainboard is reflashed with open-source field-oriented control (FOC) firmware and commanded by the Raspberry Pi over universal asynchronous receiver-transmitter (UART) (Section 2). Each critical subsystem is a standalone prototype (P1–P5) that must pass its own tests before it is integrated into the complete cart (P6). Work runs in three parallel tracks. The motion chain is serial, drive base (P1) then safety layer (P3) then golfer following (P2), because following cannot be tested until the cart can drive and be stopped. The smart bag (P4) and distance display (P5) need no motion, so they are developed on the bench and hand-carried in parallel.

The riskiest proofs of concept run first, in November and December: flashing the mainboard (DRV-1), reading radio-frequency identification (RFID) tags near metal club shafts (BAG-1), and measuring ultra-wideband (UWB) range and bearing to the golfer's tag (TRK-1, TRK-2). Safety also sets the order. Firmware speed and current limits, the serial-timeout stop, and a hardwired kill switch (DRV-2) are verified before the cart moves near anyone. The wireless e-stop (SAF-3) is verified before the first following test (TRK-3), because during following trials the safety operator is not standing at the cart.

Jp Masotti owns the motion chain: the drive base (P1) and golfer tracking and following (P2). Connor Gorman owns the smart bag (P4) and distance display (P5). Vince Song owns the safety layer (P3) and the integration hardware (power, mounting, and wiring), and leads INT-1 with Connor. The whole team carries out INT-2 to INT-4. Every member has milestones in every month (Table 4), with fewer in December because of exams.

5.2 Prototype Development Plan

Table 3 defines each prototype. Each one moves from a bench or static test to a test in its operating conditions, and P6 integration begins only after P1–P5 pass on their own. The proof-of-concept stages (DRV-1, BAG-1, TRK-1, TRK-2) come first because a failure there forces a hardware change (Section 5.4).

The club-out interlock in P3 is part of the minimum viable product (MVP). A golfer takes a club out of the bag to swing it, and a swung club reaches about 1.9 m from the golfer, which covers the following distance (H-09, Section 4). P4 already knows when a club is out, so blocking motion toward the golfer then adds no hardware.

5.3 Milestones and Dependencies

Table 4 lists the MVP milestones. Dates follow from dependencies and part lead times: the RFID, UWB, and Global Positioning System (GPS) modules are ordered the week of 2 November so that BAG-1, TRK-1, and GPS-1 can hold. Kingston grass is likely snow-covered from about December to late March, so DRV-4 runs before DRV-3, using ballast in place of the chassis, to finish grass testing in November. Following and INT-2 run indoors over the winter, and field tests return in March (C-6). Two weeks of buffer separate INT-2 from INT-3, and one week separates INT-4 from the final demonstration.

Integration order. The motion chain (P1, P3, P2) is integrated first so the cart stops reliably before anything else is added. The bag and display (P4, P5) are then mounted on that cart (INT-1), and the full system runs on flat indoor ground (INT-2) before going to the field (INT-3, INT-4). At INT-1, four compatibility checks must pass. (1) Power: the Pi 5 (5 V, 5 A), reader, GPS, display, and UWB modules run from the 36 V pack through a direct current to direct current (DC-DC) converter, with no brownout during motor acceleration. (2) Pi I/O: UARTs for the mainboard, GPS, and possibly UWB, Serial Peripheral Interface (SPI) for the reader, and Inter-Integrated Circuit (I2C) and general-purpose input/output (GPIO) for sensors and the e-stop receiver, with no pin or bus conflicts. (3) Motor noise: RFID reads, UWB ranging, and GPS fix quality hold while the hub motors run. (4) Timing: the following and stop loops keep their period with all processes running.

Stretch goals (Table 18, Appendix C) start only after the MVP milestones they depend on have passed, and are taken up by whichever member's track is ahead.

Table 3: Prototype definitions. Milestone X is complete when test T-X passes (Section 6).

Scope	Development stages (low → high fidelity)	Success criteria
P1 Drive base (DRV-1–4): speed-limited motion commanded by the Pi, with a stop independent of the Pi		
Reflashed mainboard in speed mode on a UART link; firmware speed/current limits; serial-timeout stop; kill switch; chassis with locked pivot, cradle, handle, caster	Wheels raised on bench → floor with ballast equal to a loaded bag → short grass	Speed ≤ [REQ: max speed]; stops on serial timeout and on kill switch; braking distance on grass ≤ [REQ: braking distance]
P2 Tracking and following (TRK-1–4): follow the golfer's UWB tag, not a person it sees; stop on tag loss		
Golfer's tag; cart anchors (two-anchor ranging or AoA module) for range and bearing; following controller on the Pi commanding P1	Static range and bearing vs. tape measure (PoC) → straight-line following → turns, pace changes, second person nearby	Range/bearing error ≤ [REQ: tracking accuracy]; distance within [REQ: following distance]; stops within [REQ: tag-loss stop]; does not switch to a second person
P3 Safety layer (SAF-1–4): stop for obstacles, a remote stop, link loss, or a club out of the bag		
Front ultrasonic array plus LiDAR, covering low objects; handheld wireless e-stop with link-loss timeout and no automatic restart; club-out interlock	Static detection map → e-stop with wheels raised → stop trials on the moving cart with soft obstacles	Detects [REQ: test objects] at ≥ [REQ: detection range]; stops within [REQ: reaction time]; e-stop latency ≤ [REQ: e-stop latency] at ≥ [REQ: e-stop range]; stops within [REQ: link-loss timeout]; no motion toward the golfer while a club is out
P4 Smart bag (BAG-1–3): know which clubs are in the bag with no golfer input; warn of a club left behind		
RFID reader and club tags (fallback in C-2); inventory state machine; missing-club warning logic	One tagged club in the real bag (PoC) → full bag, stationary → warning triggered by cart movement	Present/absent correct ≥ [REQ: inventory accuracy] over [REQ: cycles] cycles; delay ≤ [REQ: delay]; warns within [REQ: warning distance]
P5 Distance display (GPS-1–3): show the distance to the green with no golfer input		
GPS module; stored green coordinates for the test site; distance calculation; display of distance, inventory, and warnings	Hand-carried GPS at surveyed points → distance vs. reference → full display	Position error ≤ [REQ: GPS error]; distance error ≤ [REQ: distance accuracy]; readable at [REQ: viewing distance]
P6 Integrated ASCC (INT-1–4): all subsystems on one cart, powered by the hoverboard battery		
36 V pack → DC-DC converter → Pi and peripherals; mounting; software integration on the Pi	Stationary integration → full system indoors → field verification → validation with a golfer	All MVP requirements verified; a golfer completes a simulated hole using only the cart

5.4 Contingency Plans

Table 5 lists the technical, safety, and schedule risks that could stop a requirement being met. Each has a measurable trigger tied to a milestone and names the substitute part or

Table 4: MVP milestones. Each is complete when its test T-⟨ID⟩ passes. Dates are Fridays (Nov–Dec 2026, Jan–Mar 2027).

ID	Milestone	Depends on	Owner	Due
DRV-1	Firmware flashed; Pi spins both wheels over UART (raised)	–	Jp	13 Nov
DRV-2	Speed/current limits, serial-timeout stop, kill switch on bench	DRV-1	Jp	20 Nov
DRV-4	Speed control and braking on floor and grass, with ballast	DRV-2	Jp	27 Nov
DRV-3	Chassis complete (locked pivot, cradle, handle, caster, enclosure)	DRV-1	Jp	4 Dec
SAF-1	Static obstacle sensing, including low objects	–	Vince	27 Nov
SAF-3	Wireless e-stop with link-loss timeout, no automatic restart	DRV-2	Vince	11 Dec
SAF-2	Obstacle stop on the moving cart	DRV-3, DRV-4, SAF-1	Vince	22 Jan
SAF-4	Club-out interlock: no motion toward golfer while a club is out	BAG-2, TRK-3, INT-1	Vince	19 Feb
BAG-1	One tagged club read in the real bag (PoC, metal shafts)	–	Connor	27 Nov
BAG-2	Full-bag present/absent inventory, bag stationary	BAG-1	Connor	11 Dec
BAG-3	Missing-club warning as the cart moves away without a club	BAG-2, GPS-1	Connor	29 Jan
GPS-1	Position error at surveyed outdoor points	–	Connor	27 Nov
GPS-2	Distance to green from stored coordinates vs. reference	GPS-1	Connor	15 Jan
GPS-3	Display shows distance, inventory, and warnings	GPS-2, BAG-2	Connor	5 Feb
TRK-1	UWB range to the golfer's tag, static (PoC)	–	Jp	11 Dec
TRK-2	UWB bearing to the golfer's tag, static (PoC)	TRK-1	Jp	15 Jan
TRK-3	Straight-line following indoors; stops on tracking loss	DRV-3, DRV-4, TRK-2, SAF-3	Jp	29 Jan
TRK-4	Turns and pace changes; a second person does not take over	TRK-3	Jp	12 Feb
INT-1	Bag and display on the stationary cart, cart power; compatibility checks	DRV-3, BAG-3, GPS-3	Vince, Connor	12 Feb
INT-2	Full system running on flat indoor ground	INT-1, TRK-4, SAF-2, SAF-4	All	26 Feb
INT-3	Verification: field test of all MVP requirements, incl. one-charge range	INT-2	All	12 Mar
INT-4	Validation: a golfer plays a simulated hole using only the cart	INT-3	All	19 Mar

method. Dropping stretch goals is not a contingency, since the MVP does not depend on them.

Table 5: Contingency plans. Added cost is given in parentheses.

ID	Trigger	Change to system or approach	Milestones affected
C-1	Mainboard is a split or unsupported type, or flashing fails, at DRV-1	Use the community firmware fork for that board (split-board and AT32-based forks exist), or buy a second used hoverboard after confirming a supported single mainboard (second hoverboard)	DRV-1 slips up to 2 weeks, pushing DRV-4 toward snow (C-6)
C-2	BAG-1 read rate below [REQ: inventory accuracy] near metal shafts	On-metal NFC tags in the grip ends, one small antenna per slot through an SPI multiplexer (antennas). Last resort: per-slot presence sensing, which keeps the missing-club warning but loses club identity and needs supervisor agreement (sensors)	BAG-2 slips up to 2 weeks; BAG-3, GPS-3, INT-1 follow
C-3	TRK-2 bearing error too large to hold [REQ: following distance]	Switch bearing method between two-anchor ranging and an AoA module (second UWB or AoA module)	TRK-3; TRK-4 and INT-2 follow
C-4	GPS-1 position error above [REQ: GPS error]	Average fixes while the cart is stationary, which is when the golfer stops to hit and reads the distance; add an active antenna mounted high on the handle (antenna)	GPS-2
C-5	RFID, UWB, or GPS errors, or Pi brownouts, with motors running at INT-1	Read the bag only while stationary, since clubs are only removed then; separate DC-DC supply for the Pi and reader; ferrites and twisted motor leads; antennas moved away from the hub motors (converter, ferrites)	INT-1 slips; INT-2 buffer absorbs up to 2 weeks
C-6	Snow cover before DRV-4, INT-3, or INT-4	Grass tests done in November; following and INT-2 run indoors; INT-3/INT-4 move to a cleared paved lot or after snowmelt, with pavement-only results stated as a deviation	DRV-4, INT-3, INT-4
C-7	Wheels stall or slip on grass or slopes at DRV-4	Restrict operation to short, flat grass and state it as a terrain limitation, or switch to a larger-wheel off-road hoverboard with a confirmed supported mainboard (off-road hoverboard)	DRV-4, INT-3
C-8	Any track slips more than 2 weeks	The owner of a track that is ahead (expected: Connor after GPS-3) joins the late track; stretch work pauses	INT-2, INT-3; buffer absorbs up to 2 weeks
C-9	T-DRV-4 deceleration below [REQ: a_req] (H-01)	Lower the firmware speed cap and lengthen the minimum following distance until Eq. (1) holds, and update those requirements	DRV-4; TRK-3 runs with the new limits
C-10	T-SAF-3 latency above [REQ: e-stop latency] or link drops within [REQ: e-stop range] (H-04)	Replace the Pi-read receiver with a dedicated industrial remote e-stop receiver wired directly to the power relay (receiver)	SAF-3 slips; TRK-3 cannot start until it passes

6. Verification and Validation

6.1 Verification Strategy and Traceability

6.2 Subsystem and Prototype Tests

6.3 Experimental Test Setups

6.4 System Validation

7. Resource Requirements

The ASCC has a project budget of CAD $600. Several key components, including the Raspberry Pi, ultrasonic sensors, and golf push cart, are already owned or borrowed, reducing the overall cost. The remaining budget provides flexibility for unexpected expenses, replacements, or design changes. Additional components, including motors and GPS tracking hardware, may be obtained through the ECE Department or Ingenuity Labs. However, their estimated costs are included in the budget until their availability is confirmed. Table 6 summarizes the estimated costs.

7.1 Hardware and Equipment

Table 6: Estimated ASCC bill of materials (CAD).

Component	Qty.	Cost
Raspberry Pi 5 (existing)	1	$0
Raspberry Pi display	1	$40
Ultrasonic sensors (existing)	Multiple	$0
RFID readers and tags	2 sets	$20
TF-Luna LiDAR sensor	1	$50
GPS tracker	1	$30
Golf push cart (existing)	1	$0
Motors	1 set	$80
Motor controller	1	$50
Battery and power supply	1 set	$75
Mounting hardware	Assorted	$30
Wiring, connectors, switches	Assorted	$30
Tax and shipping	–	$75
Total		**$480**

7.2 Facilities and Contingencies

Development will require electronics tools and access to indoor and outdoor testing areas. Tools can be found utilizing the ’65 ECE lab, and testing will occur in controlled areas and hopefully on a real golf course for the finished prototype. Python and Git will support software development.

The estimated cost is $480, leaving $120 below the $600 budget limit. This remaining budget allows for additional components or design adjustments if needed.

8. Conclusion

The ASCC blueprint expands upon the initial proposal by establishing a structured system design, measurable requirements, safety analysis, and detailed development plan. The project aims to integrate golfer following, obstacle detection, emergency stopping, RFID club tracking, and GPS-based course information into a functional autonomous golf cart.

Stakeholder needs have been decomposed into functional, performance, interface, and safety requirements, each linked to planned verification tests. Quantitative targets for following distance, speed, obstacle detection, RFID accuracy, and battery runtime provide measurable acceptance criteria. The hazard analysis further identifies risks involving collisions, uncontrolled movement, electrical faults, and terrain conditions, with corresponding safeguards and residual risks documented.

Development will follow an incremental approach, prioritizing motor control, safety, and tracking before full system integration. Critical functions will be verified through subsystem testing and controlled fault injection before outdoor validation. Technical failures are addressed through defined contingency plans, including alternative tracking methods, RFID configurations, hardware replacements, and adjustments to operating limits. Remaining limitations include sensor blind spots, battery hazards, and restricted terrain suitability.

With an estimated cost of CAD $480 against the $600 budget, the project remains financially feasible. The next steps are to finalize quantitative safety thresholds, confirm component compatibility, and complete the initial hardware proofs of concept. Successful completion will be demonstrated through verified subsystem requirements and an integrated prototype capable of supporting a simulated golf hole under controlled conditions.

9. References

A. Detailed Requirements and Traceability

This appendix provides the complete ASCC requirements hierarchy, subsystem specifications, and verification traceability. Requirements are organized from stakeholder needs (SN) to system-level requirements (SYS) and subsystem requirements. Each requirement is assigned a priority and linked to a planned verification test.

A.1 Stakeholder Needs and System Requirements

Table 7: Stakeholder needs and associated system requirements.

ID	Stakeholder Need	System Requirements
SN-01	Transport golf equipment without manually pushing the cart.	SYS-01
SN-02	Prevent collisions and unintended movement.	SYS-02
SN-03	Identify clubs and notify users of missing equipment.	SYS-04
SN-04	Access course information and distances.	SYS-05
SN-05	Maintain reliable operation within project constraints.	SYS-06

Table 8: System-level requirements and classifications.

ID	System Requirement	Priority
SYS-01	The ASCC shall autonomously follow a golfer carrying the designated tracking controller.	Major
SYS-02	The ASCC shall prevent forward movement into detected obstacles within its stopping zone.	Critical
SYS-03	The ASCC shall provide an emergency-stop function that overrides autonomous movement commands.	Critical
SYS-04	The ASCC shall monitor the presence of individually tagged golf clubs.	Major
SYS-05	The ASCC shall provide available golf course information through a user interface.	Minor
SYS-06	The ASCC shall support battery-powered operation during controlled testing.	Major

A.2 Functional Subsystem Requirements

Table 9: Detailed functional subsystem requirements.

ID	Requirement	Parent
FL-01	The following subsystem shall calculate the required movement direction from valid tracking-controller information.	SYS-01
FL-02	The following subsystem shall update its movement command when the golfer changes position relative to the cart.	SYS-01
OB-01	The obstacle subsystem shall measure distances to objects in the forward detection region.	SYS-02
OB-02	The obstacle subsystem shall issue a stop request when a detected obstacle enters the stopping zone.	SYS-02
MC-01	The motor-control subsystem shall execute valid movement commands received from the main controller.	SYS-01
MC-02	The motor-control subsystem shall disable motor actuation in response to an emergency-stop request.	SYS-03
CT-01	The club-tracking subsystem shall associate detected RFID identifiers with registered golf clubs.	SYS-04
CT-02	The club-tracking subsystem shall generate a missing-club alert when a registered club remains absent beyond the configured detection interval.	SYS-04
CD-01	The course-data subsystem shall display available hole information for the selected course.	SYS-05
CD-02	The course-data subsystem shall display an estimated distance to a selected course target when valid location data is available.	SYS-05

A.3 Performance and Operational Requirements

Table 10: Detailed performance and operational requirements.

ID	Requirement	Parent
FL-03	The ASCC shall maintain a following distance between 1 and 3 m during controlled walking trials.	SYS-01
MC-03	The ASCC shall achieve a forward speed of at least 2 m/s on level terrain under the specified test load.	SYS-01
OB-03	The obstacle subsystem shall detect the designated test obstacle at a distance of at least 1 m.	SYS-02
CT-03	The RFID subsystem shall correctly identify at least 90% of registered clubs during repeated detection trials.	SYS-04
PW-01	The power subsystem shall support at least 1 hourof continuous operation under representative test conditions.	SYS-06

A.4 Interface Requirements

Table 11: Detailed interface requirements.

ID	Requirement	Parent
IR-01	The following interface shall transmit tracking information to the main controller.	SYS-01
IR-02	The obstacle sensors shall transmit distance measurements to the main controller.	SYS-02
IR-03	The main controller shall transmit movement commands to the motor-control subsystem.	SYS-01
IR-04	The RFID interface shall transmit detected tag identifiers to the main controller.	SYS-04
IR-05	The user interface shall display club-status information generated by the club-tracking subsystem.	SYS-04
IR-06	The power subsystem shall provide the required operating voltages to the computing and motor-control electronics.	SYS-06

A.5 Safety and Fault Requirements

Table 12: Detailed safety and fault requirements.

ID	Requirement	Parent
MC-04	The ASCC shall not energize its drive motors while the emergency-stop control is engaged.	SYS-03
MC-05	The ASCC shall not resume autonomous movement following an emergency stop until a manual reset is performed.	SYS-03
FL-04	The ASCC shall not continue autonomous movement when the tracking signal has been unavailable for more than 1 s.	SYS-03
OB-04	The ASCC shall not continue forward movement when an obstacle is within the validated stopping zone.	SYS-02
OB-05	The ASCC shall not initiate autonomous movement when required obstacle sensor data is unavailable or invalid.	SYS-02

A.6 Complete Requirements Traceability Matrix

Table 13: Functional and performance requirements traceability.

ID	Parent	Need	Priority	Prototype	Test
FL-01	SYS-01	SN-01	Major	P-02	VT-02
FL-02	SYS-01	SN-01	Major	P-02	VT-02
FL-03	SYS-01	SN-01	Major	P-02	VT-02
FL-04	SYS-03	SN-02	Critical	P-02	VT-08
OB-01	SYS-02	SN-02	Critical	P-03	VT-03
OB-02	SYS-02	SN-02	Critical	P-03	VT-04
OB-03	SYS-02	SN-02	Critical	P-03	VT-03
OB-04	SYS-02	SN-02	Critical	P-03	VT-04
OB-05	SYS-02	SN-02	Critical	P-03	VT-11
MC-01	SYS-01	SN-01	Major	P-01	VT-01
MC-02	SYS-03	SN-02	Critical	P-01	VT-07
MC-03	SYS-01	SN-01	Major	P-01	VT-01
MC-04	SYS-03	SN-02	Critical	P-01	VT-07
MC-05	SYS-03	SN-02	Critical	P-01	VT-07
CT-01	SYS-04	SN-03	Major	P-04	VT-05
CT-02	SYS-04	SN-03	Major	P-04	VT-05
CT-03	SYS-04	SN-03	Major	P-04	VT-05
CD-01	SYS-05	SN-04	Minor	P-05	VT-06
CD-02	SYS-05	SN-04	Minor	P-05	VT-06
PW-01	SYS-06	SN-05	Major	P-06	VT-09

Table 14: Interface requirements traceability.

ID	Parent	Need	Priority	Prototype	Test
IR-01	SYS-01	SN-01	Major	P-02	VT-10
IR-02	SYS-02	SN-02	Critical	P-03	VT-10
IR-03	SYS-01	SN-01	Major	P-01	VT-10
IR-04	SYS-04	SN-03	Major	P-04	VT-10
IR-05	SYS-04	SN-03	Major	P-05	VT-10
IR-06	SYS-06	SN-05	Major	P-06	VT-10

A.7 Verification Test Definitions

Table 15: Planned verification tests.

ID	Test	Verification Objective
VT-01	Motor movement	Verify movement commands and target speed.
VT-02	Golfer following	Measure following distance and response to golfer movement.
VT-03	Obstacle detection	Measure detection range and sensor response.
VT-04	Obstacle response	Verify stopping behaviour when obstacles enter the stopping zone.
VT-05	RFID identification	Measure identification accuracy and missing-club alerts.
VT-06	Course information	Verify hole information and displayed distances.
VT-07	Emergency stop	Verify motor inhibition and manual-reset behaviour.
VT-08	Signal loss	Verify stopping behaviour following tracking-signal loss.
VT-09	Battery runtime	Measure continuous operating duration.
VT-10	Interface integration	Verify information and power exchange between subsystems.
VT-11	Sensor fault	Verify safe response to missing or invalid obstacle measurements.

A.8 Critical Requirement Relationships

Table 16: Critical requirement relationships.

ID	Function	Logical Element	Verification
MC-04	Inhibit motor actuation	Safety interlock and motor-control subsystem	VT-07
FL-04	Detect loss of tracking	Following controller and signal monitor	VT-08
OB-04	Prevent forward movement near obstacles	Obstacle monitor and safety controller	VT-04
OB-05	Detect invalid sensor information	Sensor validation logic	VT-11

A.9 Stretch Goal Requirements

Table 17: Optional stretch goal requirements.

ID	Optional Requirement
SG-01	The ASCC may record shot locations and distances for later review.
SG-02	The ASCC may recommend golf clubs using course information and recorded shot data.
SG-03	The ASCC may provide speech-based status updates and user interaction.
SG-04	The ASCC may provide a voice functionality to provide information through sound.
SG-05	The ASCC may provide a voice speech recognition feature for commands.

B. Bow-Tie Diagram

Figure 2 links the causes of H-01, H-03, H-04, and H-09 (Table 1) to the top event through preventive controls, and the top event to its consequences through protective controls.

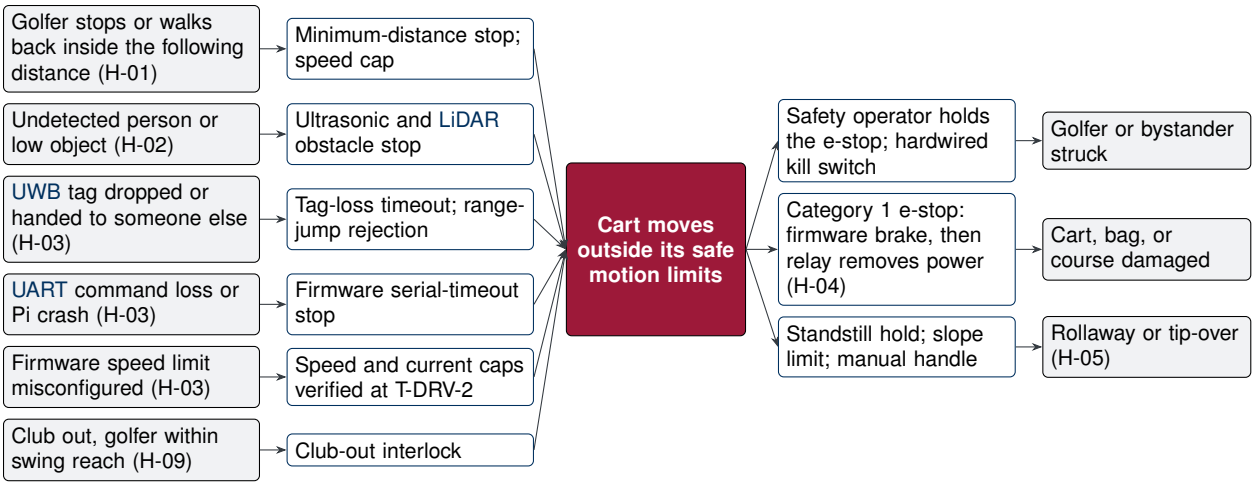


Figure 2: Bow-tie diagram for loss of motion control: causes and preventive controls (left), top event (centre), protective controls and consequences (right).

C. Stretch Milestones

Stretch milestones correspond to MAY requirements (Section 3). Each starts only after the milestones it depends on have passed (Section 5.3).

Table 18: Stretch milestones.

ID	Stretch milestone	Depends on	What makes it hard
STR-1	Voice commands: stop, follow, wait	TRK-3	Wind noise outdoors causes missed commands. Voice “stop” is an additional stop, never part of the safety function, since ISO 13850 expects a manually actuated e-stop [5]. “Stop” acts immediately; “follow” asks for confirmation
STR-2	Spoken distance callout on request	GPS-2, STR-1	Needs a speaker and microphone on the cart
STR-3	Shot logging: per-club distance from successive club-removal positions	BAG-2, GPS-1	Cart position is not ball position; club-tag trackers also miss shots [14]
STR-4	Club recommendation from logged distances	STR-3	Needs enough logged shots per club to be meaningful
STR-5	Distance from the golfer rather than the cart (cart GPS plus UWB offset)	TRK-2, GPS-2	Removes the measured-from-the-cart limitation using existing hardware
STR-6	Front/middle/back green distances; multi-hole course map	GPS-2	Needs surveyed green outlines, not single points
STR-7	Summon: cart drives to the golfer on a remote press	TRK-4	Starts beyond following range with nobody near the cart, so the obstacle stop carries all the safety
STR-8	Steer around obstacles instead of stopping	SAF-2, TRK-4	Ultrasonic sensors give a distance, not the shape of free space; needs the LiDAR
STR-9	Stay off greens and tee boxes using stored boundaries	GPS-2	Boundary margin must be at least the GPS error; wheels damage the closely mown putting surface
STR-10	Drive to the next tee from stored waypoints	GPS-2, SAF-2, STR-8	A few metres of GPS error is too much near water hazards or bunkers; likely needs RTK GPS, which is a budget question
STR-11	Slow down or stop on steep slopes using a tilt sensor	DRV-4	Addresses the rollaway and tip-over hazard (H-05); could become MVP if the team rates H-05 higher